# TODD CLASSES OF CALABI–YAU VARIETIES NEED NOT BE QUASI-EFFECTIVE

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Abstract. We construct a Calabi–Yau variety of dimension nine whose eighth Todd class is not quasi-effective, answering negatively the Calabi–Yau part of a question of Cao and Jiang.

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## 1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let $X$ be a smooth projective variety of dimension $d$ with $c_1(X) = 0 \in H^2(X, \mathbb{R})$ and $\mathcal{L}$ a nef and big line bundle on $X$. Kawamata's effective non-vanishing conjecture [Kaw00, Conjecture 2.1] predicts that $H^0(X, \mathcal{L}) \neq 0$. See also [Amb99, Section 4]. Cao and Jiang formulated this case as [CJ20, Conjecture 1.2]. For an integer $j \geq 0$, we let $\mathrm{td}_j(X)$ be the codimension $j$ part of the Todd class $\mathrm{td}(T_X)$, regarded as a cycle class on $X$ of dimension $d - j$ with rational coefficients. By Kawamata–Viehweg vanishing and the Hirzebruch–Riemann–Roch theorem,

$$h^0(X, \mathcal{L}) = \chi(X, \mathcal{L}) = \sum_{j=0}^{d} \frac{\mathrm{td}_j(X) \cdot c_1(\mathcal{L})^{d-j}}{(d-j)!}.$$

Since

$$\mathrm{td}_0(X) \cdot c_1(\mathcal{L})^d = c_1(\mathcal{L})^d > 0,$$

we have $H^0(X, \mathcal{L}) \neq 0$ when $\mathrm{td}_j(X) \cdot c_1(\mathcal{L})^{d-j} \geq 0$ for any integer $1 \leq j \leq d$.

Cao and Jiang [CJ20, Section 1] considered the following two classes of varieties with $c_1 = 0$. Let $X$ be a compact Kähler manifold of dimension $d \geq 3$. We call $X$ a *Calabi–Yau variety* if the canonical bundle of $X$ is trivial and $h^0(X, \Omega_X^i) = 0$ for any integer $0 < i < d$ (since $h^0(X, \Omega_X^2) = 0$, such $X$ is a smooth projective variety by the Kodaira embedding theorem and Chow's theorem, see Definition 2.1). A *hyperkähler variety* is a simply connected smooth projective variety $X$ such that $H^0(X, \Omega_X^2)$ is spanned by a 2-form that is non-degenerate at each point of $X$.

Cao and Jiang [CJ20, Definition 1.7] introduced the following positivity for cycle classes. Let $X$ be a smooth projective variety and $\gamma$ an algebraic cycle class on $X$ of dimension $k$ with rational coefficients. We say that $\gamma$ is *quasi-effective* if $\gamma \cdot c_1(\mathcal{L})^k \geq 0$ for any nef line bundle $\mathcal{L}$ on $X$ (see Definition 2.2). If $c_1(X) = 0 \in H^2(X, \mathbb{R})$ and $\mathrm{td}_j(X)$ is quasi-effective for any integer $1 \leq j \leq \dim X$, then [CJ20, Conjecture 1.2] holds for $X$, since each term with $j \geq 1$ in the

*Date*: November 10, 2026.

2020 *Mathematics Subject Classification.* 14J32, 14C17, 14C40, 52B20.

*Key words and phrases.* Calabi–Yau varieties, Todd classes, quasi-effective cycles, Hirzebruch–Riemann–Roch theorem, Ehrhart polynomials, double covers.

Hirzebruch–Riemann–Roch sum for $h^0(X, \mathcal{L})$ is then non-negative. They asked the following question.

**Question 1.1** ([CJ20, Question 1.8])**.** Are Todd classes of a Calabi–Yau variety or a hyperkähler variety quasi-effective?

The following cases of Question 1.1 were known. Let $X$ be a smooth projective variety of dimension $d$ with $c_1(X) = 0 \in H^2(X, \mathbb{R})$ and $j$ an integer with $0 \leq j \leq d$. Then $\mathrm{td}_j(X)$ is quasi-effective in the following cases:

(1) When $j = 2$ [Miy87, Yau77].
(2) When $X$ is a hyperkähler variety [CJ20, Jia23].
(3) When $j = 4$ and $X$ is a general complete intersection Calabi–Yau variety in a product of projective spaces such that each defining equation of $X$ has positive degree on each projective space factor [CJ20].
(4) When $d \geq 3$ and $X$ is a Wehler $d$-fold, that is, a general hypersurface of degree $(2, \ldots, 2)$ in $(\mathbb{P}^1)^{d+1}$ [FLT26].

Thus the hyperkähler part of Question 1.1 has a positive answer, and the Calabi–Yau part was open.

**Theorem 1.2** ([CLNP18, Example 1.5])**.** *Let $P \subset \mathbb{R}^9$ be the polytope defined by*

$$x_i \leq 1 \ (1 \leq i \leq 8), \quad -1 \leq x_9 \leq 1, \quad -\sum_{i=1}^{4} x_i + 4x_9 \leq 1, \quad -\sum_{i=5}^{8} x_i - 4x_9 \leq 1,$$

*where $x_1, \ldots, x_9$ are the coordinates of $\mathbb{R}^9$. Then there exists a polynomial $\mathrm{ehr}_P(m) \in \mathbb{Q}[m]$ such that $\mathrm{ehr}_P(m) = \#(mP \cap \mathbb{Z}^9)$ for any integer $m \geq 1$, and*

$$\mathrm{ehr}'_P(0) = -\frac{6673}{630}.$$

In [CLNP18], the polynomial $\mathrm{ehr}_P(m)$ is computed using polymake.

The goal of this paper is to answer the Calabi–Yau part of Question 1.1 negatively.

**Construction-Theorem 1.3.** Let $Z := \mathbb{P}^4 \times \mathbb{P}^4$. For integers $a, b$, let

$$\mathcal{O}_Z(a, b) := \mathrm{pr}_1^* \mathcal{O}_{\mathbb{P}^4}(a) \otimes \mathrm{pr}_2^* \mathcal{O}_{\mathbb{P}^4}(b),$$

where $\mathrm{pr}_1$ and $\mathrm{pr}_2$ are the first and second projections $Z \to \mathbb{P}^4$. Let $\mathcal{E} := \mathcal{O}_Z \oplus \mathcal{O}_Z(4, -4)$, let

$$Y := \mathrm{Proj}_Z \bigoplus_{k \geq 0} \mathrm{Sym}^k \mathcal{E}$$

be the projective bundle of $\mathcal{E}$, with projection $f : Y \to Z$ and tautological line bundle $\mathcal{O}_Y(1)$, and let

$$\mathcal{A} := \mathcal{O}_Y(2) \otimes f^* \mathcal{O}_Z(1, 9).$$

Then $\mathcal{O}_Y(K_Y) \cong \mathcal{A}^{-1}$, $\mathcal{A}$ is very ample, and there exists $s \in H^0(Y, \mathcal{A}^{\otimes 2})$ whose zero divisor is smooth.

For any such $s$, let $X := \mathrm{Spec}_Y(\mathcal{O}_Y \oplus \mathcal{A}^{-1})$, where the multiplication on the summand $\mathcal{A}^{-1}$ is given by $s : \mathcal{A}^{\otimes(-2)} \to \mathcal{O}_Y$, let $\pi : X \to Y$ be the projection, and let $\mathcal{L} := \pi^* \mathcal{A}$. Then $X$ is a Calabi–Yau variety of dimension 9, $\mathcal{L}$ is ample, and

$$\mathrm{td}_8(X) \cdot c_1(\mathcal{L}) = -\frac{6673}{315}.$$

In particular, $\mathrm{td}_8(X)$ is not quasi-effective.

**Remark 1.4.** Since $\mathcal{A}$ is very ample and $\pi$ is surjective, $H^0(X, \mathcal{L}) \supset \pi^* H^0(Y, \mathcal{A}) \neq 0$. Thus $X$ and the ample line bundle $\mathcal{L}$ in Construction-Theorem 1.3 satisfy the conclusion of [CJ20, Conjecture 1.2].

**Sketch of the proof.** By Hirzebruch–Riemann–Roch, the number $\mathrm{td}_8(X)\cdot c_1(\mathcal{L})$ is the coefficient of $m$ in the polynomial $\chi(X,\mathcal{L}^{\otimes m})$. Since $\mathcal{O}_Y(K_Y)\cong\mathcal{A}^{-1}$ by the relative Euler sequence of $f$ (Step 1 of the proof in Section 2) and $\pi_*\mathcal{O}_X=\mathcal{O}_Y\oplus\mathcal{A}^{-1}$, the projection formula and Serre duality on $Y$, where $\dim Y=9$, imply

$$\chi(X,\mathcal{L}^{\otimes m})=\chi(Y,\mathcal{A}^{\otimes m})-\chi(Y,\mathcal{A}^{\otimes(-m)})\quad\text{for any integer } m$$

(Step 5). Thus $\mathrm{td}_8(X)\cdot c_1(\mathcal{L})$ is twice the coefficient of $m$ in the polynomial $\chi(Y,\mathcal{A}^{\otimes m})$.

In Step 4, we prove that

$$\chi(Y,\mathcal{A}^{\otimes m})=\mathrm{ehr}_P(m)\quad\text{for any integer } m.$$

Since both sides are polynomials in $m$, it suffices to prove

$$\chi(Y,\mathcal{A}^{\otimes m})=\sum_{j=-m}^{m}\binom{5m-4j+4}{4}\binom{5m+4j+4}{4}=\#(mP\cap\mathbb{Z}^9)\quad\text{for any integer } m\geq 0;$$

the first equality follows from $R^qf_*\mathcal{A}^{\otimes m}=0$ for any integer $q>0$ and

$$f_*\mathcal{A}^{\otimes m}\cong\bigoplus_{j=-m}^{m}\mathcal{O}_Z(5m-4j,5m+4j),$$

and the second from counting the points of $mP\cap\mathbb{Z}^9$ with $x_9=j$ after the substitution $u_k=m-x_k$, $v_k=m-x_{k+4}$ $(1\leq k\leq 4)$. Thus Theorem 1.2 implies

$$\mathrm{td}_8(X)\cdot c_1(\mathcal{L})=2\,\mathrm{ehr}_P'(0)=-\frac{6673}{315}.$$

In Step 2, we show that $\mathcal{A}$ is very ample, so $s$ exists by Bertini. In Step 3, we prove that $X$ is a Calabi–Yau variety.

**Acknowledgements.** The work was partially supported by the National Key R&D Program of China #2024YFA1014400. Jihao Liu would like to thank Bin Dong, Guoxiong Gao, Zeming Sun, Bin Wu, and Leheng Chen for AI assistance. Jihao Liu would like to thank Ruochuan Liu and Gang Tian for consistent support and encouragement.

## 2. Proof of the main theorem

**Definition 2.1** ([CJ20, Section 1])**.** Let $X$ be a compact Kähler manifold of dimension $d\geq 3$. We say that $X$ is a *Calabi–Yau variety* if the canonical bundle of $X$ is trivial and $h^0(X,\Omega_X^i)=0$ for any integer $0<i<d$ (since $h^0(X,\Omega_X^2)=0$, such $X$ is a smooth projective variety by the Kodaira embedding theorem and Chow's theorem).

**Definition 2.2** ([CJ20, Definition 1.7])**.** Let $X$ be a smooth projective variety and $\gamma$ an algebraic cycle class on $X$ of dimension $k$ with rational coefficients. We say that $\gamma$ is *quasi-effective* if $\gamma\cdot c_1(\mathcal{L})^k\geq 0$ for any nef line bundle $\mathcal{L}$ on $X$.

*Proof of Construction-Theorem 1.3.* **Step 1.** Since $\det\mathcal{E}\cong\mathcal{O}_Z(4,-4)$, $\mathcal{O}_Z(K_Z)\cong\mathcal{O}_Z(-5,-5)$, and $\omega_{Y/Z}\cong\mathcal{O}_Y(-2)\otimes f^*\det\mathcal{E}$, we have

$$\mathcal{O}_Y(K_Y)\cong\mathcal{O}_Y(-2)\otimes f^*\mathcal{O}_Z(-1,-9)\cong\mathcal{A}^{-1}.$$

**Step 2.** Let $\mathcal{M}:=\mathcal{O}_Y(1)\otimes f^*\mathcal{O}_Z(0,4)$. Since

$$\mathcal{E}\otimes\mathcal{O}_Z(0,4)\cong\mathcal{O}_Z(0,4)\oplus\mathcal{O}_Z(4,0)$$

is globally generated, $\mathcal{M}$ is globally generated and $\mathcal{M}\otimes f^*\mathcal{O}_Z(1,1)$ is very ample, so

$$\mathcal{A}\cong\mathcal{M}\otimes\big(\mathcal{M}\otimes f^*\mathcal{O}_Z(1,1)\big)$$

is very ample. By Bertini, there exists $s\in H^0(Y,\mathcal{A}^{\otimes 2})$ whose zero divisor is smooth.

**Step 3.** In this step we prove that $X$ is a Calabi–Yau variety of dimension 9, that $\mathcal{L}$ is ample, and that

$$\pi_*\mathcal{O}_X \cong \mathcal{O}_Y \oplus \mathcal{O}_Y(K_Y).$$

Since the zero divisor of $s$ is smooth, $X$ is smooth. Since $\pi$ is finite and surjective and $Y$ is projective of dimension 9, $X$ is projective of dimension 9, and $\mathcal{L} = \pi^*\mathcal{A}$ is ample. Since $\pi$ is the double cover determined by $\mathcal{A}$ and branched along the smooth zero divisor of $s \in H^0(Y, \mathcal{A}^{\otimes 2})$, by [BHPV04, Lemma I.17.1(iii)] and Step 1,

$$\mathcal{O}_X(K_X) \cong \pi^*\big(\mathcal{O}_Y(K_Y) \otimes \mathcal{A}\big) \cong \mathcal{O}_X.$$

By the definition of $X$ and Step 1,

$$\pi_*\mathcal{O}_X = \mathcal{O}_Y \oplus \mathcal{A}^{-1} \cong \mathcal{O}_Y \oplus \mathcal{O}_Y(K_Y).$$

Since $\mathcal{O}_Y \cong \mathcal{O}_Y(K_Y) \otimes \mathcal{A}$ and $\mathcal{A}$ is ample, Kodaira vanishing implies $h^q(Y, \mathcal{O}_Y) = 0$ for any integer $q > 0$. By Serre duality, $h^q(Y, \mathcal{O}_Y(K_Y)) = h^{9-q}(Y, \mathcal{O}_Y)$ for any integer $0 \le q \le 9$. Since $\pi$ is finite, for any integer $0 \le q \le 9$,

$$h^q(X, \mathcal{O}_X) = h^q(Y, \mathcal{O}_Y) + h^{9-q}(Y, \mathcal{O}_Y) = \begin{cases} 1 & \text{if } q \in \{0, 9\}, \\ 0 & \text{if } 1 \le q \le 8. \end{cases}$$

Since $h^0(X, \mathcal{O}_X) = 1$, $X$ is connected. Thus $X$ is a connected smooth projective variety of dimension 9, hence a compact Kähler manifold of dimension 9. By the Hodge decomposition and Hodge symmetry [GH78, p. 116], for any integer $1 \le q \le 8$,

$$h^0(X, \Omega_X^q) = h^q(X, \mathcal{O}_X) = 0.$$

Thus $X$ is a Calabi–Yau variety.

**Step 4.** In this step we prove that

$$\chi(Y, \mathcal{A}^{\otimes m}) = \mathrm{ehr}_P(m) \quad \text{for any integer } m.$$

Let $m \ge 0$ be an integer. Since

$$\mathcal{A}^{\otimes m} \cong \mathcal{O}_Y(2m) \otimes f^*\mathcal{O}_Z(m, 9m) \quad \text{and} \quad \mathrm{Sym}^{2m}\mathcal{E} \cong \bigoplus_{i=0}^{2m} \mathcal{O}_Z(4i, -4i),$$

we have

$$f_*\mathcal{A}^{\otimes m} \cong \mathrm{Sym}^{2m}\mathcal{E} \otimes \mathcal{O}_Z(m, 9m) \cong \bigoplus_{i=0}^{2m} \mathcal{O}_Z(m + 4i, 9m - 4i),$$

and $R^q f_*\mathcal{A}^{\otimes m} = 0$ for any integer $q > 0$. For any integer $0 \le i \le 2m$, $m + 4i \ge 0$ and $9m - 4i \ge m \ge 0$. By Leray, Künneth, and the cohomology of $\mathcal{O}_{\mathbb{P}^4}(a)$ for $a \ge 0$, $h^q(Y, \mathcal{A}^{\otimes m}) = 0$ for any integer $q > 0$, and

$$\chi(Y, \mathcal{A}^{\otimes m}) = \sum_{i=0}^{2m} \binom{m + 4i + 4}{4} \binom{9m - 4i + 4}{4} = \sum_{j=-m}^{m} \binom{5m - 4j + 4}{4} \binom{5m + 4j + 4}{4}, \tag{2.1}$$

where the second equality is the substitution $j = m - i$.

In the coordinates $x_1, \dots, x_9$ of Theorem 1.2, a point $x \in \mathbb{Z}^9$ belongs to $mP$ if and only if

$$x_k \le m \ (1 \le k \le 8), \quad -m \le x_9 \le m, \quad -\sum_{k=1}^{4} x_k + 4x_9 \le m, \quad -\sum_{k=5}^{8} x_k - 4x_9 \le m.$$

Let $j$ be an integer such that $-m \le j \le m$. For $x \in \mathbb{Z}^9$ with $x_9 = j$, let $u_k := m - x_k$ and $v_k := m - x_{k+4}$ for $1 \le k \le 4$. Then $x \in mP$ if and only if

$$u_k \ge 0, \ v_k \ge 0 \ (1 \le k \le 4), \quad \sum_{k=1}^{4} u_k \le 5m - 4j, \quad \sum_{k=1}^{4} v_k \le 5m + 4j.$$

Since $5m - 4j \geq m \geq 0$, $5m + 4j \geq m \geq 0$, and

$$\#\Big\{(u_1, \dots, u_4) \in \mathbb{Z}^4_{\geq 0} \;\Big|\; \sum_{k=1}^{4} u_k \leq t\Big\} = \binom{t+4}{4} \quad \text{for any integer } t \geq 0,$$

we have

$$\#\{x \in mP \cap \mathbb{Z}^9 \mid x_9 = j\} = \binom{5m-4j+4}{4}\binom{5m+4j+4}{4}.$$

Since $-m \leq x_9 \leq m$ for any $x \in mP$, (2.1) implies

$$\chi(Y, \mathcal{A}^{\otimes m}) = \#(mP \cap \mathbb{Z}^9) \quad \text{for any integer } m \geq 0.$$

By Hirzebruch–Riemann–Roch, there exists $Q \in \mathbb{Q}[m]$ such that $Q(m) = \chi(Y, \mathcal{A}^{\otimes m})$ for any integer $m$. Since $Q(m) = \chi(Y, \mathcal{A}^{\otimes m}) = \#(mP \cap \mathbb{Z}^9)$ for any integer $m \geq 0$, Theorem 1.2 implies $Q(m) = \operatorname{ehr}_P(m)$ for any integer $m \geq 1$, so $Q = \operatorname{ehr}_P$. Thus $\chi(Y, \mathcal{A}^{\otimes m}) = \operatorname{ehr}_P(m)$ for any integer $m$.

**Step 5.** In this step we prove that

$$\operatorname{td}_8(X) \cdot c_1(\mathcal{L}) = -\frac{6673}{315}.$$

Let $m$ be an integer. Since $\pi$ is finite, by the projection formula and Step 3,

$$\pi_* \mathcal{L}^{\otimes m} \cong \mathcal{A}^{\otimes m} \oplus \big(\mathcal{A}^{\otimes m} \otimes \mathcal{O}_Y(K_Y)\big),$$

so, since $\dim Y = 9$, Serre duality implies

$$\chi(X, \mathcal{L}^{\otimes m}) = \chi(Y, \mathcal{A}^{\otimes m}) - \chi(Y, \mathcal{A}^{\otimes(-m)}).$$

Thus, by Step 4,

$$\chi(X, \mathcal{L}^{\otimes m}) = \operatorname{ehr}_P(m) - \operatorname{ehr}_P(-m) \quad \text{for any integer } m. \tag{2.2}$$

On the other hand, by Hirzebruch–Riemann–Roch on $X$,

$$\chi(X, \mathcal{L}^{\otimes m}) = \sum_{n=0}^{9} \frac{m^{9-n}}{(9-n)!} \operatorname{td}_n(X) \cdot c_1(\mathcal{L})^{9-n} \quad \text{for any integer } m. \tag{2.3}$$

The right-hand sides of (2.2) and (2.3) are polynomials in $m$ whose value at any integer $m$ is $\chi(X, \mathcal{L}^{\otimes m})$, so they are equal. The coefficient of $m$ is $2\operatorname{ehr}'_P(0)$ in (2.2) and $\operatorname{td}_8(X) \cdot c_1(\mathcal{L})$ in (2.3). By Theorem 1.2,

$$\operatorname{td}_8(X) \cdot c_1(\mathcal{L}) = 2\operatorname{ehr}'_P(0) = -\frac{6673}{315}.$$

Since $\mathcal{L}$ is nef, $\operatorname{td}_8(X)$ is a cycle class of dimension 1, and $\operatorname{td}_8(X) \cdot c_1(\mathcal{L}) < 0$, $\operatorname{td}_8(X)$ is not quasi-effective.

The theorem follows. □

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